



CFD simulation of the impact of tip clearance on the hydrothermal performance and entropy generation of a water-cooled pin-fin heat sink

Amin Shahsavari^{a,*}, Mohammad Shahmohammadi^a, Ighball Baniasad Askari^{b,*}

^a Department of Mechanical Engineering, Kermanshah University of Technology, Kermanshah, Iran

^b Department of Mechanical Engineering, faculty of Engineering, University of Zabol, Zabol, Iran

ARTICLE INFO

Keywords:

Convection heat transfer
Entropy generation analysis
Hydrothermal performance
Pin-fin heat sink
Tip clearance

ABSTRACT

In this paper, the 3D numerical calculations of the water flow inside a pin-fin heat sink (PFHS) to determine the influence of tip clearance on the hydrothermal behavior and frictional & thermal irreversibilities of the fluid flow. The numerical simulations were conducted for five different fin heights ($\xi = 1.5$ mm to 2.5 mm) considering four different Reynolds numbers ($Re=500$, 1000, 1500, and 2000). The results demonstrated that the highest heat transfer coefficient is associated with $\xi = 2.25$ mm due to more uniform vorticity distributions, while the minimum central processing unit (CPU) average temperature is obtained for $\xi = 2.5$ mm due to the higher heat transfer surface area. The minimum thermal and frictional entropy generation rates are associated with PFHS with $\xi = 1.5$ mm. However, the PFHS with $\xi = 2.25$ mm has the highest Performance Evaluation Criterion (PEC) of nearly equal to 1.28 as compared to the other ξ s. Besides, the fluid flow from the pin-fin base to the top and vice versa leads to create the vortices especially at the entrance of the PFHSs with tip clearance.

1. Introduction

The Electronic Devices (EDs) needs proper heat dissipation to decrease their temperature during their longtime usage. The effective solutions are those which entail heat transfer improvement at lower and lower pumping power while keeping the space and cost constraints. In this regard, the application of pin or plate finned heat sinks (HSs) in the miniaturized EDs has been proposed by the many researchers. The PFHS provides the extended effective surface area, a proper fluid mixing, and delaying of the viscous thermal boundary layer which causes the heat transfer augmentation [1,2]. The previous research studies revealed that there are several factors that affect the hydrothermal performance of the PFHSs such as the fluid flow rate inside the HS, the flow distribution at the inlet and outlet of the headers, the fin material, arrangement, spacing, porosity, morphology, and Tip Clearance (TC) [3–7].

The TC is a distance between the fin tip and HS cover plate which was firstly introduced by Sparrow et al. [8,9]. The TC can reduce or enhance the heat transfer coefficient inside the HS. In fact, part of the fluid flows above the pin tip which leads to decrease the fluid velocity around the pin-fins, and consequently, the heat transfer on the pin-fin surface is decreased. On the other hand, the pin-fin tip surface area entails additional heat transfer for the HS cooling process. In addition, the fluid

mixing and flow three dimensionality are enhanced by making the TC [10]. Based on a study performed by Wirtz et al. [11], the bypass flow in the TC region diminishes the heat transfer coefficient. Thereby, the optimization of TC gap helps to increase the heat transfer and flow mixing while decreasing the bypass flow adverse effect. The following literature addresses the previous researches that have been conducted on the application of TC in the PFHS.

Different TC gaps in a PFHS were examined by Mahajan [12]. The results showed that the pressure drop penalty and heat transfer gain are reduced with an increase in the TC gap. Therefore, the authors concluded that the minimum TC height is suitable for maintaining the HS thermal performance. J.Y. Min et al. [13] investigated the influence of TC on cooling performance of a PFHS for a fixed pumping power. The findings indicated that the application of TC is useful when the TC gap is smaller than the channel width.

The fluid flow inside a finned HS with TC gaps was experimentally investigated by Sastre et al. [14], where it was shown that the generated Nusselt-Graetz curves don't collapsed in a single fit and distinct heat transfer rate is obtained for different fin walls depending on the TC geometry. Zhigang et al. [15] investigated the effect of using TC in a micro-cylinders-group HS for low Nusselt numbers (Nu) of 8 to 400 and two different arrangements. They observed that the TC has low effect on the flow resistance and Nu in low Re numbers. Also, the heat transfer

* Corresponding authors.

E-mail addresses: a.shahsavari@kut.ac.ir (A. Shahsavari), ighball.baniasad@uoaz.ac.ir (I.B. Askari).

<https://doi.org/10.1016/j.icheatmasstransfer.2021.105400>

Nomenclature			
C_p	Specific heat capacity (J/kg.K)	θ	Temperature uniformity (K.m ² /W)
D_h	Hydraulic diameter (m)	μ	Viscosity (kg m ⁻¹ s ⁻¹)
h	Convective Heat transfer coefficient (W/m ² .K)	ρ	Density (kg m ⁻³)
k	Thermal conductivity (W/m.K)	ξ	Fin height (mm)
L	Length (m)	Subscripts	
P	Pressure (Pa)	f	Fluid
PEC	Performance evaluation criterion	in	inlet
q''	Heat flux (W/m ²)	m	mean
R	Thermal resistance (K.m ² /W)	max	Maximum
Re	Reynolds number	min	Minimum
$\dot{S}_{fr}$	Frictional entropy generation rate (W/K)	o	outlet
$\dot{S}_{th}$	Thermal entropy generation rate (W/K)	s	Solid
T	Temperature (K)	Abbreviations	
$\vec{V}$	Velocity vector (m/s)	CPU	Central processing unit
$\dot{V}$	Volumetric flow rate (m ³ /s)	ED	Electronic device
$\dot{W}_p$	Pumping power (W)	HS	Heat sink
Greek symbols		PFHS	Pin-fin heat sink
		TC	Tip clearance

performance is firstly increased steeply with increasing the Re , however, it is diminished at higher Re numbers. Rajabi Far et al. [16] studied the application of TC in a HS with beveled fins and phase change material slurry as the coolant. They concluded that the heat transfer efficiency is improved if an optimum TC gap is exerted. Based on the results of Dogan et al. [17], the total heat transfer rate is escalated as TC is reduced; especially, in the larger Re numbers. The experimental simulations of the forced convection heat transfer of a finned plate performed by Kim et al. [18] and Min et al. [13] revealed that the heat transfer rate increases as the flow velocity near the fin tips is gradually increased. Conversely, the heat transfer rate is decreased as the bypass flow near the TC region increases. A group of numerical and experimental studies was applied by Park and Chung [19] to investigate the heat transfer behavior of the fluid flow inside a square duct with finned plates. Based on the results, the optimum TC increases the heat transfer rate to 33% higher than that is obtained for the largest TC. Also, the influence of TC on the heat transfer rate is diminished by increasing the fin spacing. The heat transfer and pressure drop in a pin-fin microchannel was investigated by Reyes et al. [20] considering three TCs of 250 mm, 500 mm, and 1000 mm. Their results demonstrated that the heat transfer rate and pumping power of the configuration with optimum TC are respectively 83% and 18% of the base configuration without TC. Mei et al. [21] performed a group of numerical and experimental analyses to determine the influence of using TC in a micro reactor with micro pin-fin arrays. Based on that study, the introduction of TC diminishes the pressure drop and increases the heat transfer rate at low Re numbers. Also, new correlations were proposed for the heat transfer and friction factor accounting for the TC effects. The same results were obtained by Jadhav and Balaji [22], and Ho et al. [23], and they concluded that the effect of TC is more prominent for lower flow velocities. The effect of using TC on the hydrothermal performance of the shrouded pin-fins was experimentally investigated by Moores and Joshi [24] for the Re of between 200 and 10,000. The results demonstrated that for the height to diameter ratios of between 0.5 and 1.1, the mean heat transfer coefficient is increased by 15 and 23%, respectively.

The second law analysis is useful to evaluate the energy losses and irreversibilities of a heat transfer process. Furthermore, it is possible to assess the local irreversible losses at each point of the computational region through the entropy generation analysis. A great number of the previous research studies are associated with the second law analysis of HSs. For example, Shen et al. [25] used the entropy generation analysis to assess the heat transfer performance of a novel X-structured double-

layered HS. Also, the entropy analysis was applied by Ambreen et al. [26] to study the cooling performance of a nanofluid inside an aluminum foam HS. In a study performed by Alfaryjat et al. [27], the effect of Re and heat flux on the irreversibility behavior of aqueous magnetite nanofluid inside a hexagon HS were investigated. Shahsavari et al. [28] investigated the influence of using splitter in a plate PFHS using the first and second law analysis. The local thermal and frictional irreversibilities inside the HS was determined using the entropy generation analysis. In a research conducted by Al-Rashed et al. [29], an entropy generation analysis was performed to evaluate the thermal and frictional irreversibilities of a nanofluid flow inside a wavy HS.

As can be seen from the literature, the first law has been vastly used in the performance analysis of PFHSs with tip clearance. However, there has been only one contribution on the entropy generation analysis of the TC PFHSs. Only in the research performed by Shi and Dong [30], the entropy generation analysis was used to investigate the convective heat transfer of the fluid flow inside a staggered array TC pin-fin microchannel. They performed an entropy analysis to determine the optimal combination of different effective parameters on the HS performance. They reported that the thermal entropy generation rate is higher than that of the frictional term. Also, it was concluded that the TC effect at high and low aspect ratios (height to diameter) of the pin-fin structure more pronounced with the conductive effects and convective nature, respectively. Since the influence of TC on the thermal and hydrothermal irreversibilities of PFHS has been rarely studied using the entropy generation analysis, in the present study the effect of the TC gap on the hydrothermal behavior and thermal and frictional entropy generation rates of the water flow inside a PFHS were conducted for the first time. The 3-D numerical analysis of the laminar forced convection of the pure water flow inside a microchannel with TC pin-fin was conducted using the ANSYS Fluent 18.1 considering five different ξ of 1.5 mm, 1.75 mm, 2 mm, 2.25 mm, and 2.5 mm at four Re numbers of 500, 1000, 1500, and 2000. There are several computational fluid dynamics (CFD) approaches for fluid flow and heat transfer simulations such as Finite Difference ([31]), Finite Element ([32,33]), Finite Volume ([34]), and Lattice Boltzmann Method ([34–37]). In the present study, the finite volume method was used for the numerical calculation by using the ANSYS FLUENT software.

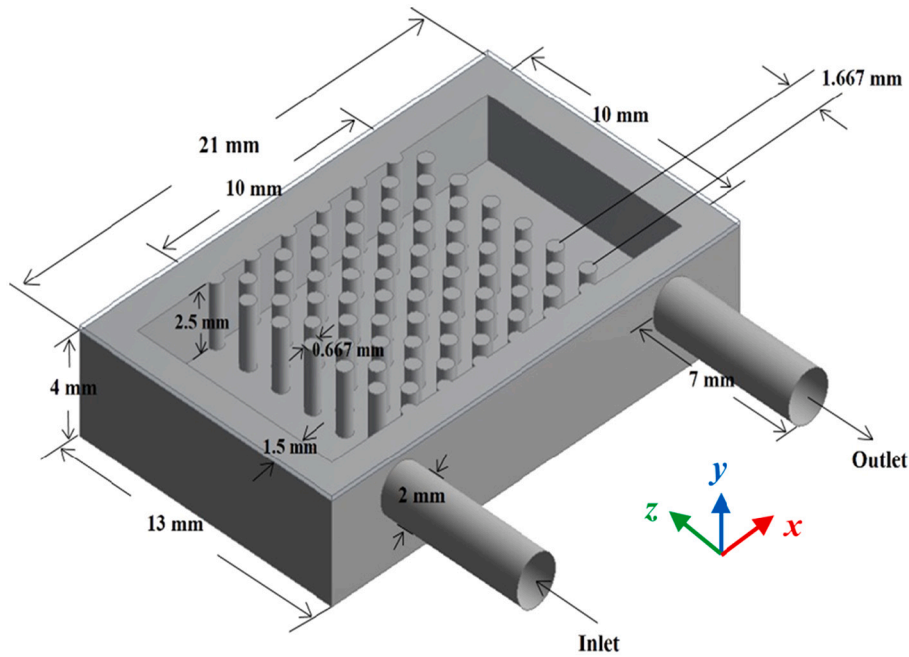


Fig. 1. Schematics of the PFHS under evaluation in this contribution.

Table 1
Outcomes of mesh independence study.

Grid size	$T_{CPU, max}$ (K)	T_{out} (K)
450,512	311.37	304.65
901,024	309.44	303.92
1,802,048	307.39	301.12
2,592,188	307.23	301.05

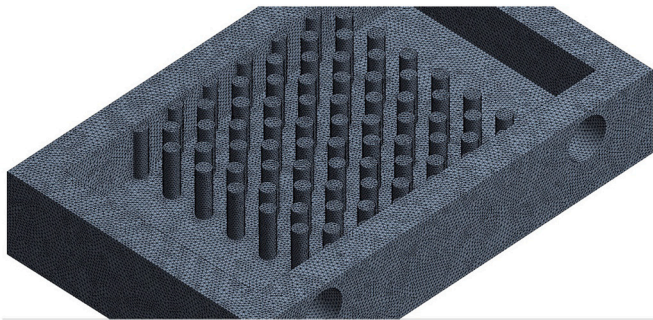


Fig. 2. Zoomed view of the grid topology.

Table 2
Findings of verification.

Δp (Pa)	R (K. m ² /W)			$\dot{V}$ (mL/min)		
	Present study	Chiu et al. [39]	Error (%)	Present study	Chiu et al. [39]	Error (%)
689	0.432	0.473	-8.7	98	104	-5.8
1370	0.384	0.417	-7.9	135	145	-6.9
2040	0.345	0.373	-7.5	178	186	-4.3
2750	0.307	0.331	-7.3	212	224	-5.4

2. Geometrical configuration of models and boundary conditions

Fig. 1 shows the geometry of TC PFHS under study in this work, which is made of copper and placed on the top of a CPU to cool it down. A constant heat flux of $q'' = 80000 \text{ W/m}^2$ is exerted on the bottom plate of the PFHS while the other walls are assumed to be adiabatic. The working fluid is pure water with density, thermal conductivity, specific heat, and viscosity of 998.1 kg/m^3 , 0.6 W/m.K , 4.162 kJ/kg.K , and 0.001003 kg/m.s , respectively, which enters the PFHS at constant temperature of 300 K . The fluid is assumed to be at the liquid state and thereby, the single phase flow is studied. It is assumed that the fluid with laminar regime, steady conditions, liquid phase, and constant characteristics incompressibly flows through the PFHS.

3. Governing equations

The hydrothermal behavior of incompressible fluid flow inside the PFHS was investigated by using the mass, momentum and energy balances which leads to obtain the pressure (P), velocity and temperature (T) fields inside the numerical domain at three dimensions (x, y, z) [28]:

Mass conservation:

$$\nabla \cdot \vec{V} = 0 \quad (1)$$

Momentum:

$$\rho (\vec{V} \cdot \nabla) \vec{V} = -\nabla P + \rho \vec{g} + \mu \nabla^2 \vec{V} \quad (2)$$

Energy conservation for fluid:

$$\rho C_p (\vec{V} \cdot \nabla) T = k_f \nabla^2 T \quad (3)$$

Energy conservation for solid:

$$k_s \nabla^2 T = 0 \quad (4)$$

In above equations, k , μ , and C_p are thermal conductivity, dynamic viscosity and specific heat capacity, respectively. Also, the subscripts s and f represent the solid and fluid, respectively. The numerical calculations were performed at four different Re numbers of 500, 1000, 1500,

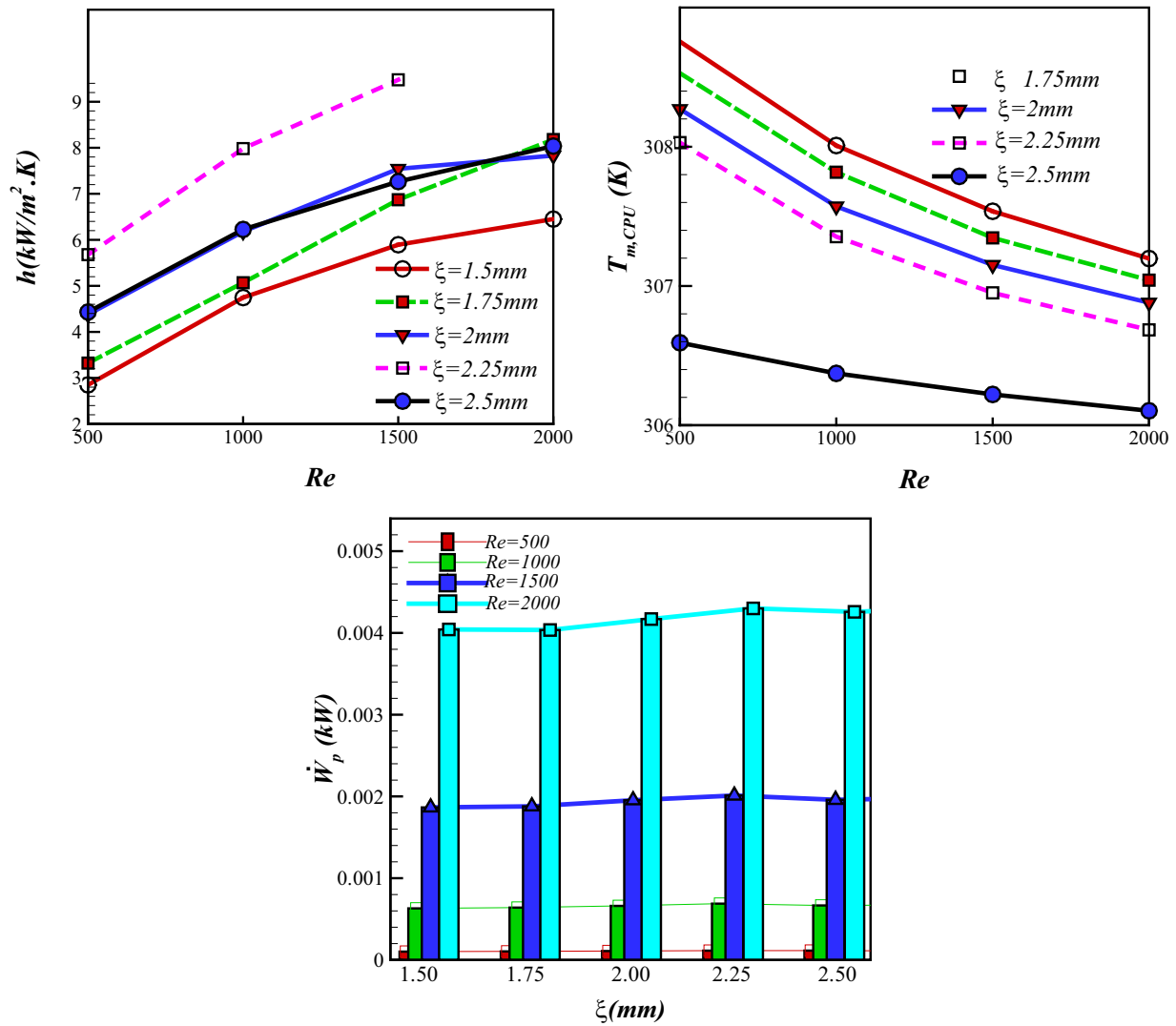


Fig. 3. a. The variations of h versus Re for different ξ s. b. The variations of average CPU temperature versus Re for different ξ s. c. The variations of pumping power versus ξ for different Re numbers.

and 2000, which are calculated using the inlet velocity (u), dynamic viscosity, and hydraulic diameter ($D_h = 4A_c L/A_t$) [28]:

$$Re = \frac{\rho u D_h}{\mu} \quad (5)$$

where, A_c is the frontal surface area, A_t is the total heat transfer surface area in the middle part of the computational domain, and L is the length of the computational domain.

The uniformity of temperature in the computational domain could be determined using the following expression [38]:

$$\theta = \frac{T_{CPU,max} - T_{CPU,min}}{\bar{q}} \quad (6)$$

where, the nominator and denominator of the above equation are the maximum temperature difference of CPU and the applied heat flux, respectively [38]. It may be noted that the higher the efficiency of the HS the lower the ratio of θ . The efficiency of the HS also could be evaluated using the thermal resistance of the HS (R) which is the ratio of the difference between the CPU mean temperature ($T_{m,CPU}$) and fluid temperature at the HS inlet (T_{in}) to the applied heat flux calculated as follows [38]:

$$R = \frac{T_{CPU,m} - T_{in}}{\bar{q}} \quad (7)$$

It is worth mentioning that at higher HS efficiencies, $T_{CPU,m}$ and consequently, R are reduced based on the assumptions of constant T_{in} and $\bar{q}$. The pumping power ($\dot{W}_p$) and convective heat transfer coefficient (h) are obtained using the following Eqs. [38]:

$$\dot{W}_p = \dot{V} \Delta p \quad (8)$$

$$h = \frac{\bar{q}}{T_{CPU,m} - T_m} \quad (9)$$

In the above formulations, Δp , $\dot{V}$, and T_m are the pressure drop, fluid volumetric flow, and average fluid temperature, respectively. Also, T_m is calculated as follows [38]:

$$T_m = \frac{T_{in} + T_{out}}{2} \quad (10)$$

The Performance Evaluation Criterion (PEC) is used to evaluate the improvement in hydrothermal behavior of the fluid flow by comparing the heat transfer and pressure drop of the coolant fluid in the PFHS with ξ of less than 2.5 mm with those inside the PFHS with $\xi = 2.5$ mm:

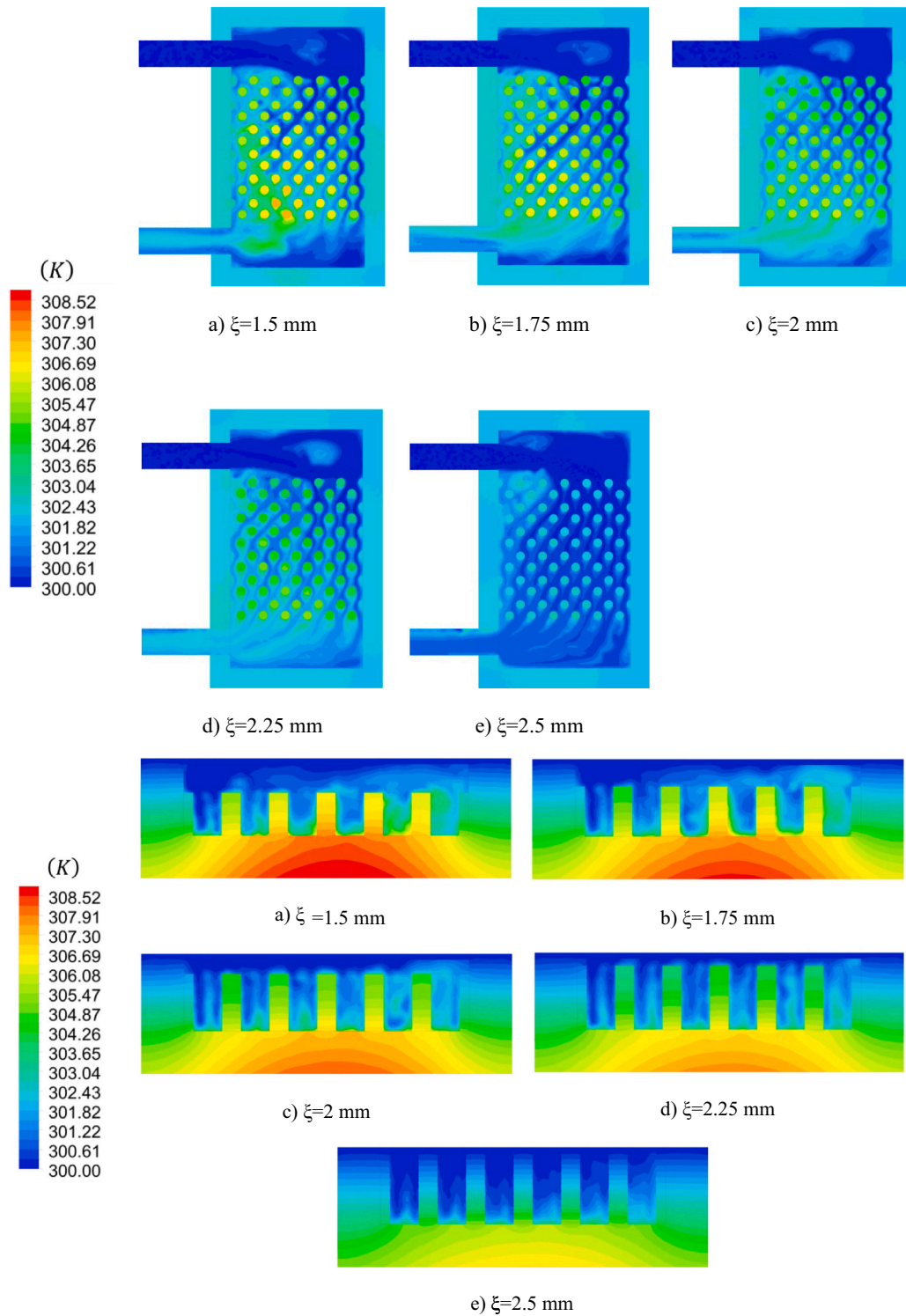


Fig. 4. a. The temperature contours along a plane (x-z) for $Re=2000$ and considering five ξ s. b. The temperature contours along a plane (z-y) for $Re=2000$ and considering five ξ s. c. The temperature contours along a plane (x-y) for $Re=2000$ and considering five ξ s.

$$PEC = \frac{h_{\xi < 2.5 \text{ mm}} / h_{\xi = 2.5 \text{ mm}}}{(f_{\xi < 2.5 \text{ mm}} / f_{\xi = 2.5 \text{ mm}})} \quad (11)$$

The irreversibilities inside the computational domain is determine by calculating the local frictional and thermal entropy generation rates ($\dot{S}_{fr}'$ and $\dot{S}_{th}'$) as well as global entropy generation rates ($\dot{S}_{fr}$ and $\dot{S}_{th}$) [38]:

$$\dot{S}_{fr}' = \frac{\mu}{T} \left\{ 2 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right] + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right)^2 + \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right)^2 \right\} \quad (12)$$

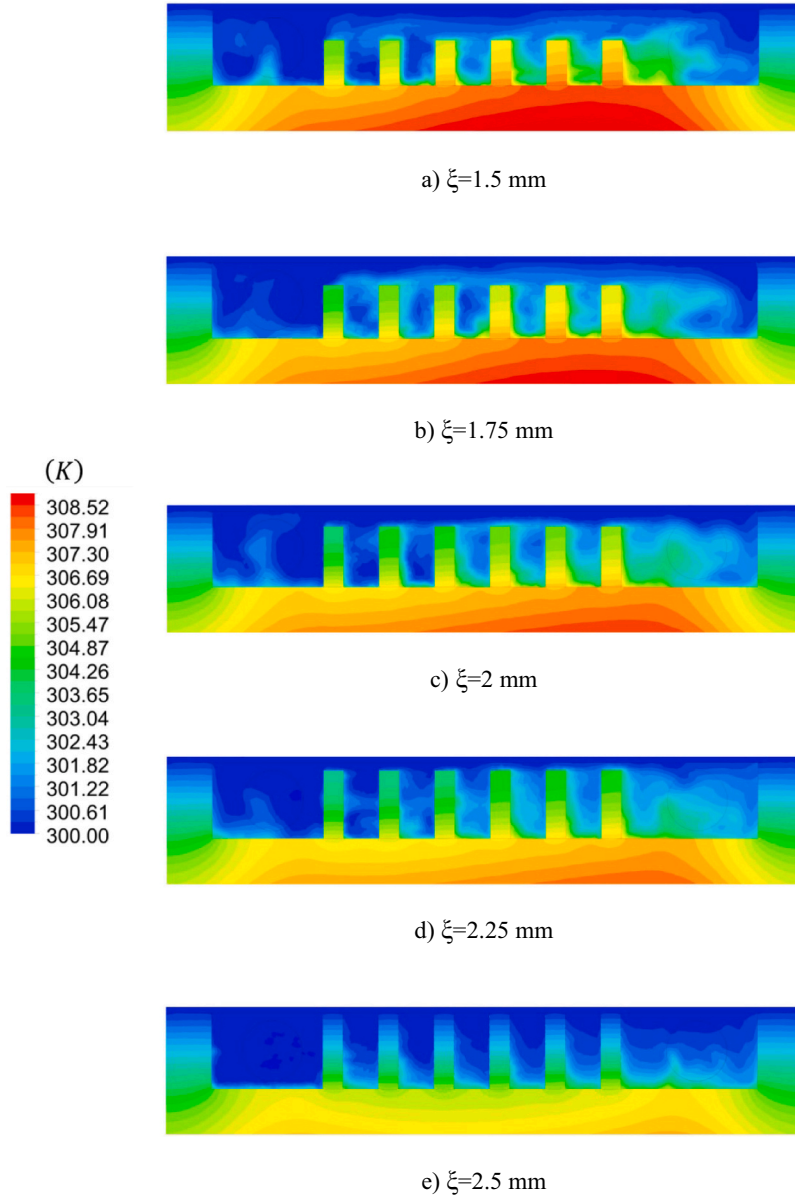


Fig. 4. (continued).

$$\dot{S}_{fr} = \int \dot{S}_{fr}' dV \quad (13)$$

$$\dot{S}_{th}' = \frac{k}{T^2} \left(\left(\frac{\partial T}{\partial x} \right)^2 + \left(\frac{\partial T}{\partial y} \right)^2 + \left(\frac{\partial T}{\partial z} \right)^2 \right) \quad (14)$$

$$\dot{S}_{th} = \int \dot{S}_{th}' dV \quad (15)$$

4. Numerical technique and verification

ANSYS Fluent 18.1 was used to perform the numerical analysis based on the finite volume method applying the second order upwind method for the momentum and energy conservation equations. The coupling between the velocity and pressure fields was applied using the SIMPLE scheme considering the convergence criterion of 10^{-6} for the governing eqs.

A finer mesh grid was considered near the walls due to higher intensity of velocity and temperature gradients. Also, the non-dependency

of the numerical results to the number of grid nodes was examined performing a numerical analysis on four grids with different node numbers to determine the suitable number of nodes. The numerical calculations were performed on the water flow at $Re = 2000$ inside the numerical domain. As can be seen from the results listed in Table 1, the increase in the number of nodes more than 1,802,048 has no considerable effect on the maximum CPU temperature ($T_{max, CPU}$) and the water temperature at the outlet of the PFHS (T_{out}). Fig. 2 shows the zoomed view of the numerical domain with the applied mesh grid.

To examine the verify the numerical technique, the results of the present numerical analysis obtained for the volumetric flow rate and thermal resistance inside a micro PFHS were compared with those are reported in Chiu et al. [39]. A good agreement between the results can be observed based on Table 2.

5. Results and discussions

The variations of convective heat transfer coefficient (h) versus Re for different ξ s are depicted in Fig. 3a. As can be seen, h is enhanced as Re

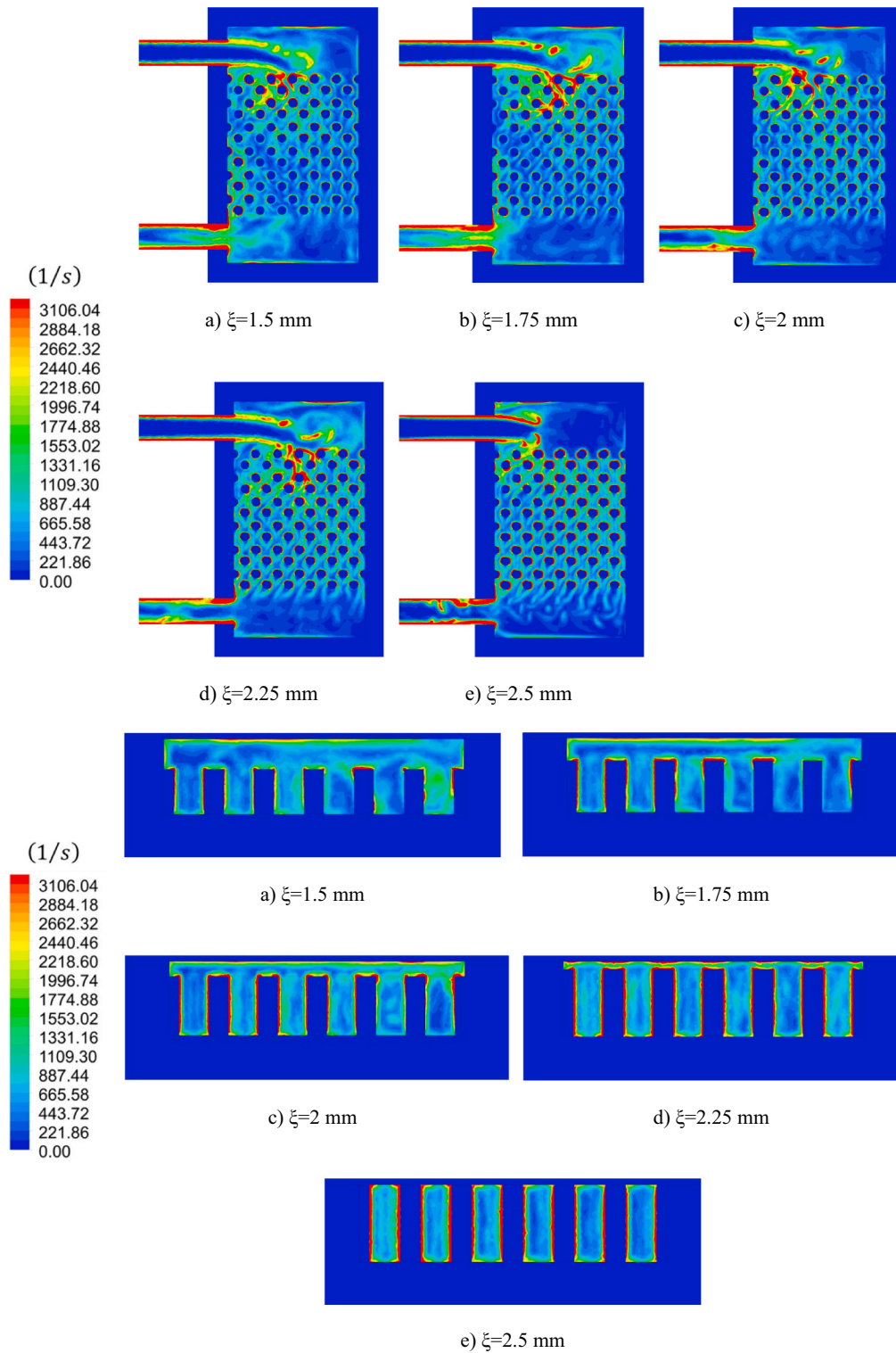


Fig. 5. a. The vorticity contours along a plane ($x-z$) for $Re=2000$ and considering five ξ s. b. The vorticity contours along a plane ($z-y$) for $Re=2000$ and considering five ξ s. c. The vorticity contours along a plane ($x-y$) for $Re=2000$ and considering five ξ s.

increases from 500 to 2000 for all ξ s. The increasing percentage of h with Re is more pronounced for lower ξ ; nearly 130% for ξ of 1.5 mm and 1.75 mm, also, 80% for ξ of 2 mm, 2.25 mm and 2.5 mm. Fig. 3a also demonstrates that the increase of ξ from 1.5 mm to 2.25 mm leads to a significant increase in h . While, at ξ of 2.5 mm, h decreases to that is obtained for ξ of 2 mm.

The influence of ξ and Re on the $T_{m, CPU}$ is depicted in Fig. 3b, where it can be seen that the minimum $T_{m, CPU}$ is obtained for $\xi = 2.5$ mm which is in contrast with that is shown in Fig. 3a, where the highest h is obtained for $\xi = 2.25$ mm. The reason can be illustrated using the convection heat transfer phenomenon which is a function of heat transfer surface area (A), h and the temperature difference between fluid and

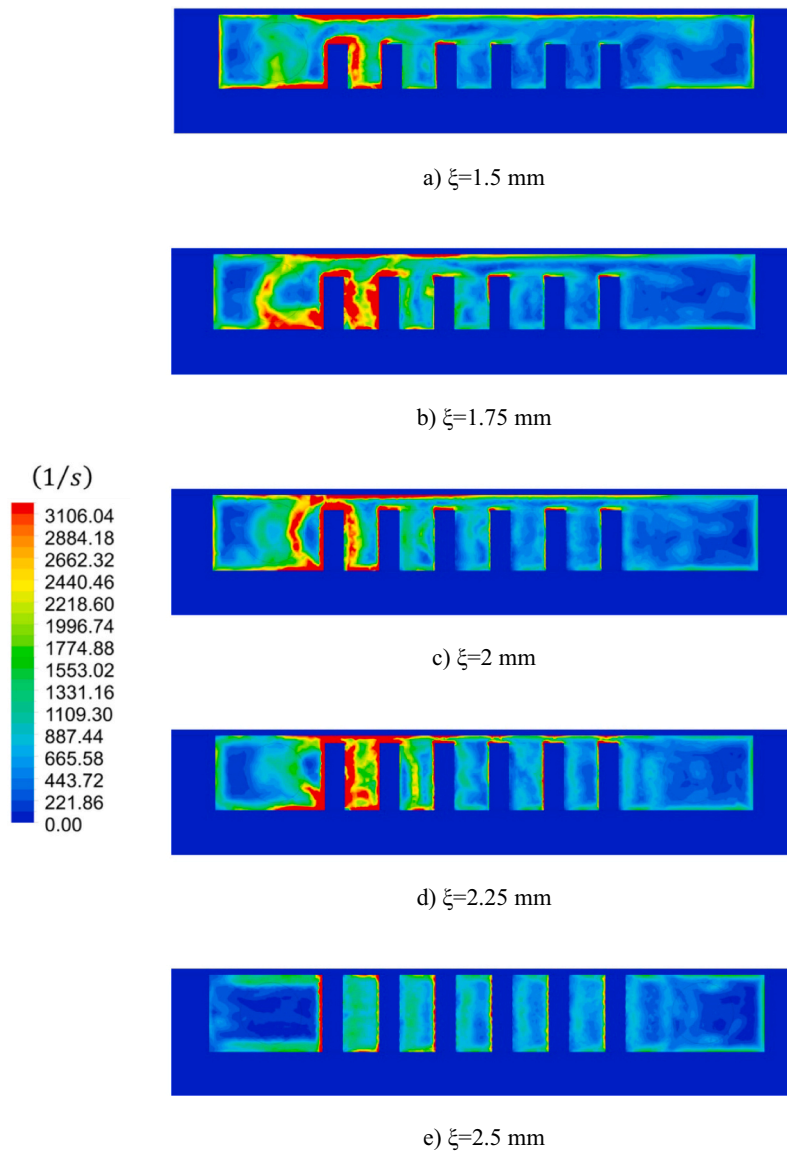


Fig. 5. (continued).

PFHS wall. In fact, A is increased as ξ increases from 2.25 mm to 2.5 mm. Conversely, the highest h is obtained for $\xi = 2.25$ mm (Fig. 3a) due to the better vorticity distributions over the fin surfaces. However, the influence of higher A in ξ of 2.5 mm dominates the effect of h in ξ of 2.25 mm, thereby the pin-fin with $\xi = 2.5$ mm has better performance in decreasing $T_{m, CPU}$ than that with $\xi = 2.25$ mm as can be seen in Fig. 3b.

Fig. 3c illustrates the variations of the fluid required pumping power ($\dot{W}_p$) versus ξ for different Re numbers. Here it can be seen that the pumping power escalates as Re is increased from 500 to 2000. Also, the increase of ξ from 1.5 mm to 2.25 mm leads to escalate the pumping power by 11.6%, 9.11%, 7.78%, and 6.36% for Re of 500, 1000, 1500, and 2000, respectively. While the further increase in ξ from 2.25 mm to 2.5 mm results in an insignificant increase in the pumping power; less than 3.25%, 2.67%, and 0.98% for Re numbers of 1000, 1500, and 2000, respectively.

Fig. 4 shows the temperature contours inside the numerical domain along three different planes (Fig. 4a (x-z), Fig. 4b (z-y), Fig. 4c (x-y)), for $Re=2000$, and five different ξ s. As it is clear in Fig. 4, the temperature contours for $\xi = 2.5$ mm contains only the cyan and blue colors due to the lower PFHS temperature. However, higher color contrast can be observed in the temperature contours for lower values of ξ . For instance,

the yellow and orange colors on the pin-fins near the PFHS outlet for $\xi = 1.5$ mm depicts the higher temperatures at this region. Also, in the low ξ s, the coolant fastly passes along the PFHS length at top of the pin-fins and reaches the outlet region without proper mixing and removing the heat from the pin-fins surfaces. That is why the lower temperature region is obtained near the lateral PFHS walls at the entrance and exit as can be seen in Fig. 4a. The temperature contours are shown in Fig. 4b for z-y plane of the PFHS lateral wall near the outlet. Here, it can be seen that for the lower ξ s the pin-fins, the temperature is higher due to the fact that a considerable part of the coolant fluid passes from top of the pin-fins without moving over the pin-fin bodies, thereby the temperature rises at this regions. For higher ξ s (2.25 mm and 2.5 mm) with the smaller tip clearances, a better distribution of the coolant fluid inside the pin-fin spaces is obtained; thereby lower temperatures over the pin-fin surfaces can be achieved. The temperature contours on the x-y plane is shown in Fig. 4c. As can be seen, the high-temperature zone is smallest for the PFHS with ξ s of 2.25 mm and 2.5 mm due to the reason that was discussed in the explanations of Fig. 4b. In fact, the coolant fluid flow within the pin-fins spaces is more uniform for the lower tip clearances (higher ξ s). As a consequence, lower temperature zone is obtained for ξ s of 2.25 mm and 2.5 mm.

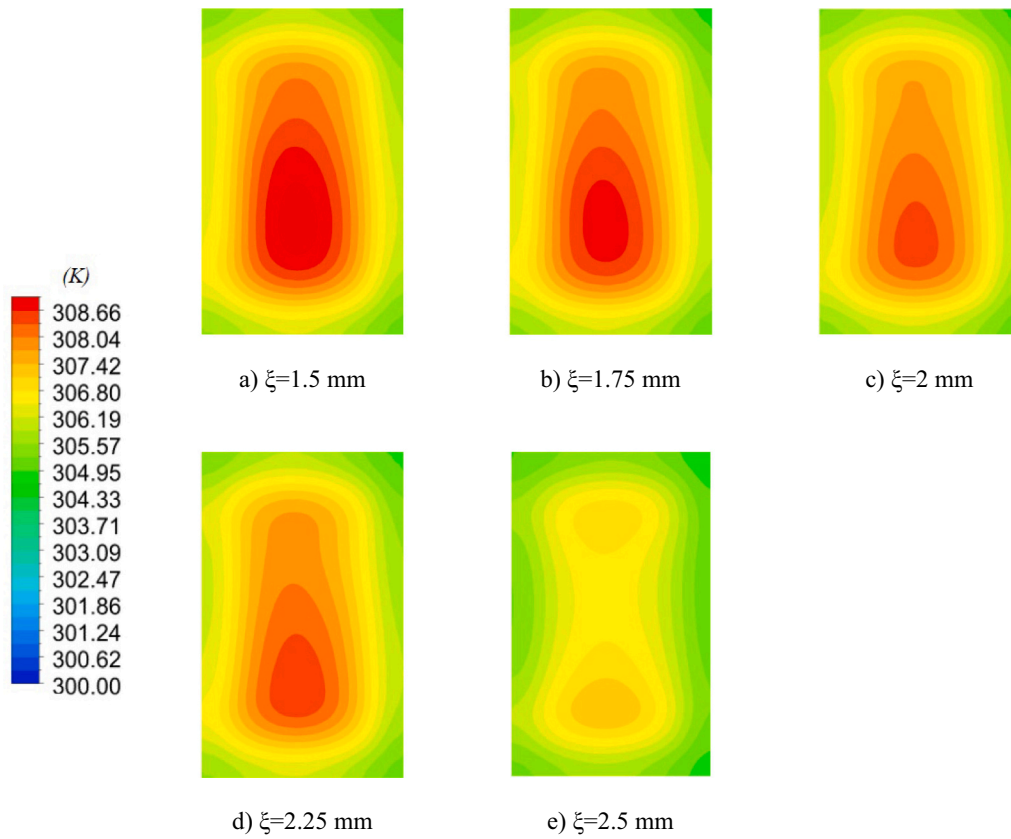


Fig. 6. The contours of CPU surface temperature for $Re=2000$ and considering five ξ s.

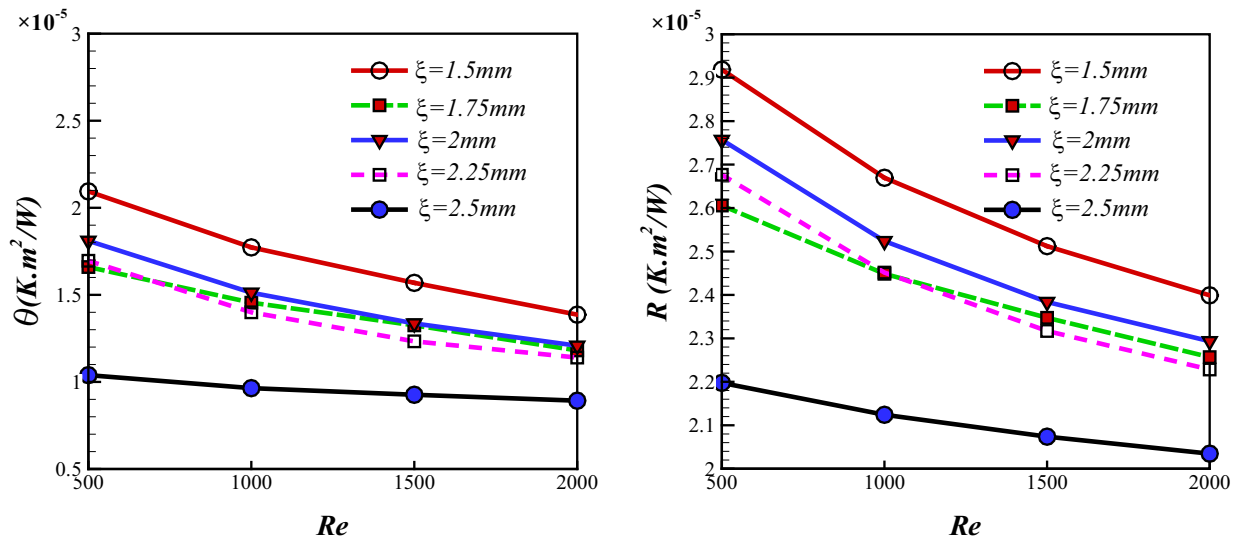


Fig. 7. a. The variations of θ versus Re for different ξ s. b. The variations of R versus Re for different ξ s.

The vorticity contours of the fluid flow at three different planes are shown in Fig. 5 for different ξ s and $Re=2000$. As can be seen from Fig. 5a (x-z plane), with the exception for ξ of 2.25 mm, the vortices are more strong at the right side of the x-z plane for other ξ s. Also, a better vorticity penetration inside the first rows of the pin-fin spaces which are closer to the inlet is obtained for the lower ξ s; more specifically in ξ s of 1.75 mm, 2 mm and 2.25 mm. For the ξ of 2.5 mm, the vortices can only distribute over the pin-fin walls due to the zero tip clearance distance. However, the use of tip clearance causes to create further vortices at pin-fin in addition to the pin-fins walls as can be clearly seen from

Fig. 5b and Fig. 5c. The higher the ξ is, the stronger vortices are obtained at the tips. It is clearly evident from Figs. 5b and 5c that the vorticity distributions inside the pin-fin spaces and at the tips are more uniform for the PFHS with $\xi = 2.25$ mm which leads to a higher heat transfer coefficient for this PFHS (Fig. 3a).

Fig. 6 discloses the contours of CPU surface temperature for different ξ s and $Re=2000$. Clearly, increasing the ξ reduces the CPU temperature and also makes the CPU temperature distribution more uniform.

As it was discussed earlier in the explanations of eqs. (8) and (9), θ and R are two parameters to evaluate the thermal performance of PFHSs.

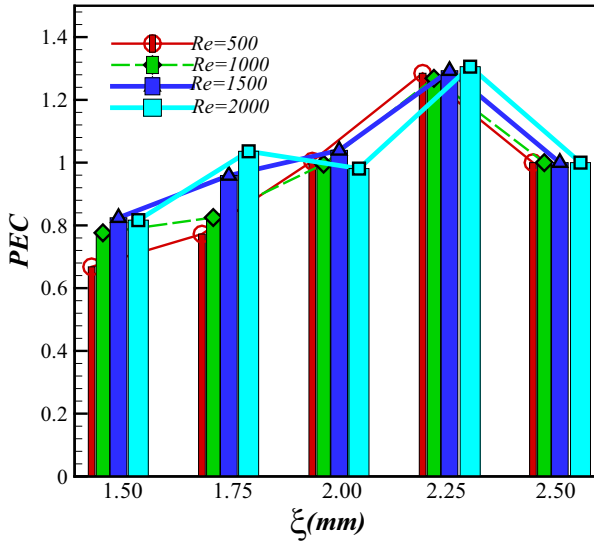


Fig. 8. The variations PEC factor versus ξ for different Re numbers.

Fig. 7 shows the variations of these two parameters versus Re for different ξ . Based on Fig. 7, the increase of Re leads to a decrease in θ and R which means the increasing of PFHS heat transfer performance, which is due to more uniform mixing of the fluid flow inside the computational geometry. Besides, the highest and lowest values of θ and R are associated with ξ of 1.5 mm and 2.5 mm, respectively. In fact, the performances of the PFHS (θ) increases as the maximum CPU temperature is decreased considering a constant heat flux on the PFHS base. Also, it is clear from Fig. 7a that for low ξ s, the influence of increasing Re on decreasing θ is high and this will be decreased by increasing ξ . For example, as Re increases from 500 to 2000, θ decreases by 51% and 16% for the PFHS with ξ of 1.5 mm and 2.5 mm, respectively. The same trend is obtained for decreasing of R with Re and, the increasing of Re from 500 to 2000 leads to decrease R by 21% and 8% for ξ s of 1.5 mm and 2.5 mm, respectively. Furthermore, the increasing of ξ from 1.5 mm to 2.25 mm (33% increase in ξ) leads to decrease θ and R by about 21.64% and 7.66%, respectively, for $Re=2000$. For higher increase in the ξ from 2.25 mm to 2.5 mm, θ and R decrease by about 27.72% and 9.51%, respectively, for the same Re of equal to 2000 based on Fig. 7.

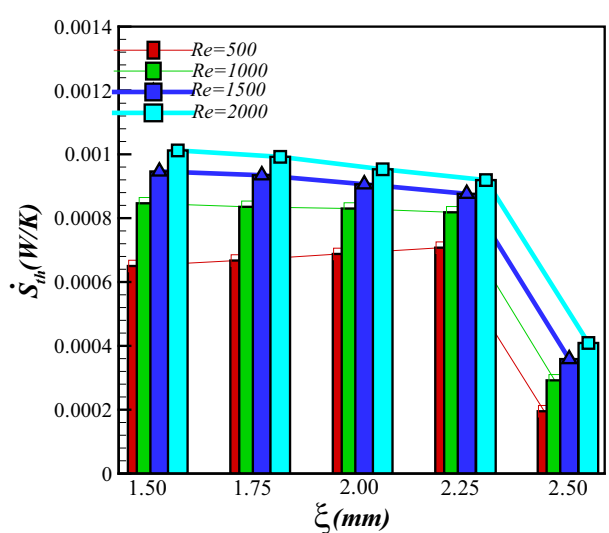
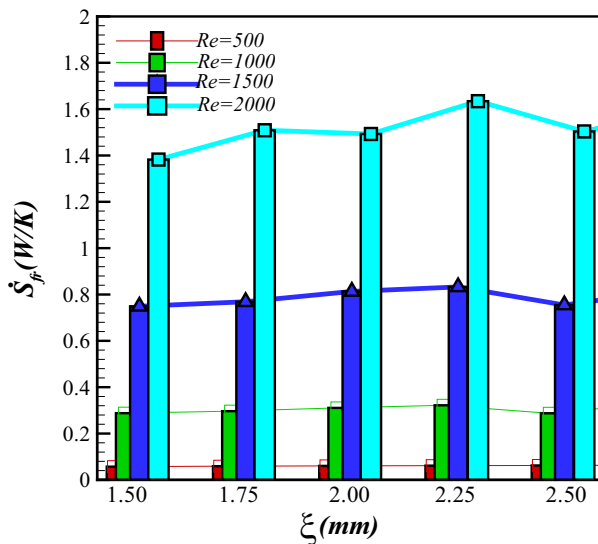


Fig. 9. a. The variations of frictional entropy generation rate versus ξ for different Re numbers. b. The variations of thermal entropy generation rate versus ξ for different Re numbers.

The PEC factor which is calculated using eq. (13) is obtained for different ξ and Re and results are shown in Fig. 8. Here, it is evident that the highest and lowest PECs are associated with $\xi = 2.25$ mm ($PEC \approx 1.3$) and $\xi = 1.5$ mm ($PEC \approx 0.7$), respectively, for different Re numbers. For ξ of 1.75 mm, 2 mm and 2.5 mm, the PEC is almost equal to 1 at different Re s. The results of Fig. 8 also show that the pin-fin has its optimum performance in $\xi = 2.25$ mm from the first law of thermodynamics viewpoint.

The variations of frictional and thermal entropy generation rates ($\dot{S}_{fr}$ and $\dot{S}_{th}$) with ξ at different Re numbers are shown in Fig. 9a and Fig. 9b, respectively. Based on Fig. 9a, the highest and lowest $\dot{S}_{fr}$ are associated with ξ of 2.25 mm and 1.5 mm, respectively, at different Re numbers. The results show that the pattern of $\dot{S}_{fr}$ changes with ξ is different in various Re numbers. At $Re=500$, an increase in ξ leads to an increase in $\dot{S}_{fr}$, while at higher Re numbers, increasing in ξ from 1.5 mm to 2.25 mm leads to an increase in $\dot{S}_{fr}$, and a further increase in ξ leads to a decrease in $\dot{S}_{fr}$. According to eq. (12), the $\dot{S}_{fr}$ is a function of viscosity, temperature and velocity gradient of water. The results show that increasing the ξ from 1.5 mm to 2.25 mm leads to a decrease in fluid temperature and, therefore, an increase in $\dot{S}_{fr}$, while the opposite occurs with increasing the ξ from 2.25 mm to 3 mm. As mentioned earlier, increasing the ξ to 2.25 mm leads to amplification of the vortices, which increases the velocity gradient and, consequently, increases the $\dot{S}_{fr}$, while increasing the ξ to 2.5 mm, due to the elimination of TC, leads to a decrease in the strength of the vortices and thus reduces the $\dot{S}_{fr}$. In addition, Fig. 9a depicts that the $\dot{S}_{fr}$ rises with the increase of Re , which is because of the decrease of fluid temperature by boosting the Re .

Fig. 9b show that at $Re=500$, the $\dot{S}_{th}$ elevates with the increase of ξ from 1.5 mm to 2.25 mm and then declines with further increase of ξ . At higher Re numbers (i.e. $Re \geq 1000$), the $\dot{S}_{th}$ diminishes with increase of ξ . According to eq. (14), the $\dot{S}_{th}$ depends on the thermal conductivity, temperature and temperature gradient of water. The temperature of the fluid decreases with increasing the ξ from 1.5 mm to 2.25 mm, which leads to an increase in the $\dot{S}_{th}$, while the opposite occurs with increasing the ξ from 2.25 mm to 2.5 mm. On the other hand, increasing the ξ from 1.5 mm to 2.25 mm reduces the temperature gradient and thus the $\dot{S}_{th}$ by improving the fluid mixing by amplifying the vortices, while at $\xi=2.5$ mm, the temperature gradient increases again, which leads to an increase in the $\dot{S}_{th}$. Furthermore, Fig. 9b illustrates that the $\dot{S}_{th}$ elevates

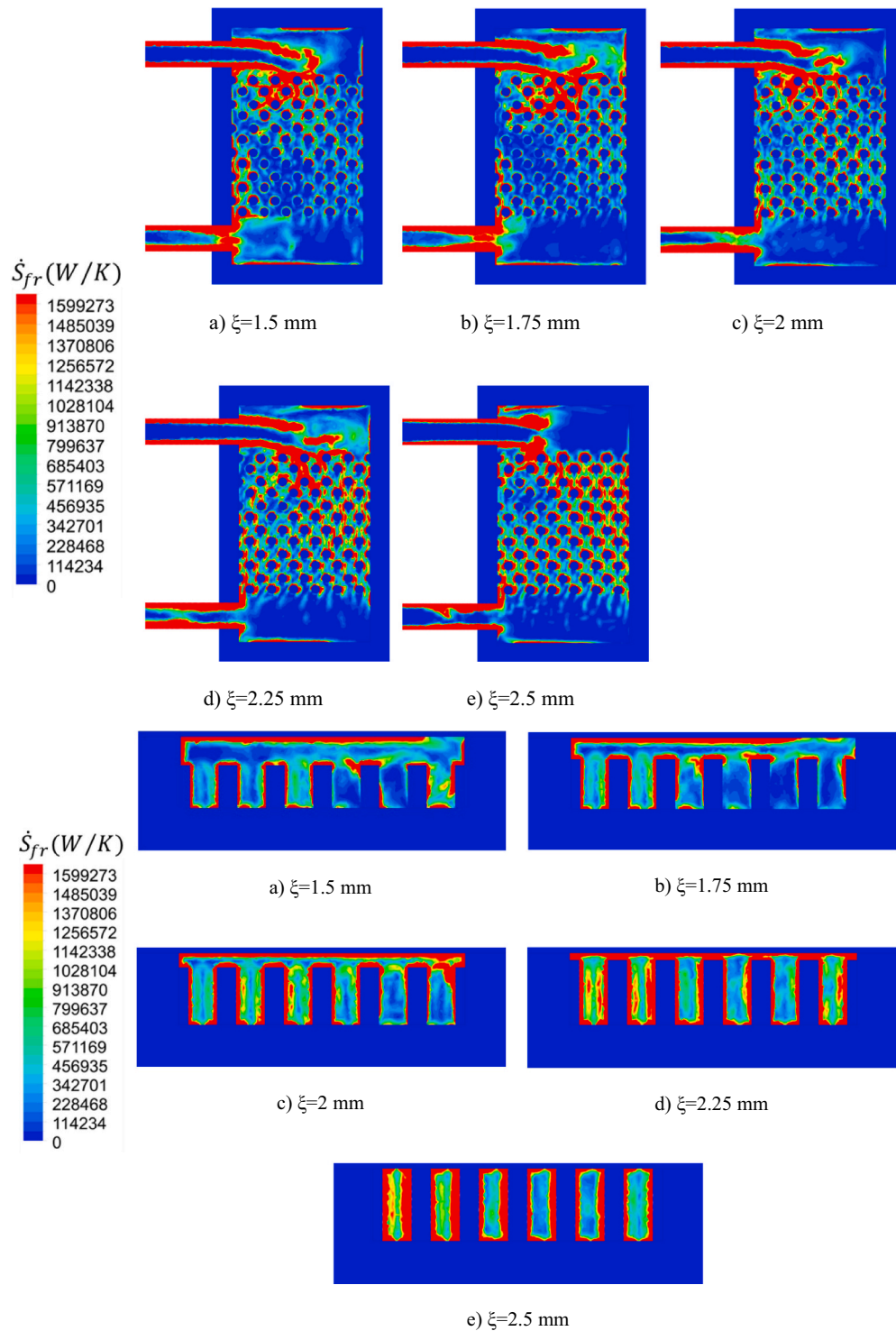


Fig. 10. a. The frictional entropy generation contours along a plane (x-z) for $Re = 2000$ and considering five ξ s. b. The frictional entropy generation contours along a plane (z-y) for $Re = 2000$ and considering five ξ s. c. The frictional entropy generation contours along a plane (x-y) for $Re = 2000$ and considering five ξ s.

with the rise of ξ , which is originated from the reduction of fluid temperature by boosting the Re . It should be noted that in all cases studied in the present study, the $\dot{S}_{fr}$ is higher than the $\dot{S}_{th}$, which makes the total entropy generation rate follow the $\dot{S}_{fr}$.

The contours shown in Fig. 10 are associated with the frictional entropy generation rate for three different plane of the computational domain. Based on Fig. 10a, for the lower ξ , the fluid fastly passes from

the tip regions at the right side of the PFHS x-z plane, while a vortex region is obtained at the left side and middle part of the PFHS on the x-z plane, more specifically near the PFHS inlet. The created vortexes and the resultant $\dot{S}_{fr}$ are more intensified in the PFHSs with ξ s of 1.75 mm to 2.25 mm as can be seen from Fig. 10c. Besides, with the increase in ξ from 1.5 mm to 2.25 mm, the vorticity distribution over the pin-fin bodies and tip clearance region is intensified, thereby $\dot{S}_{fr}$ is increased

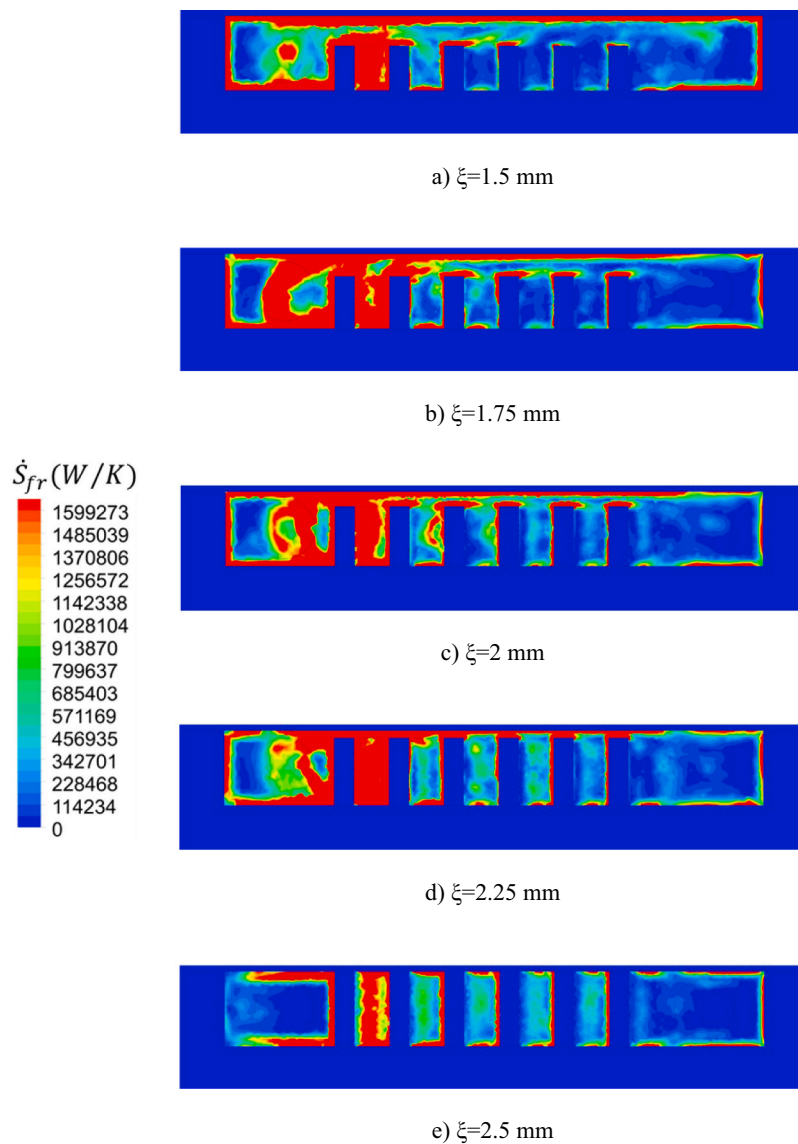


Fig. 10. (continued).

at these regions. As it was mentioned earlier in the explanations of Fig. 10a, the highest $\dot{S}_{fr}$ is associated with $\xi = 2.25$ mm which is nearly 8% higher than $\dot{S}_{fr}$ obtained for the ξ s of 2 mm and 2.5 mm.

6. Conclusions

The 3D numerical analysis of the water flow inside a PFHS was conducted at different Re and five ξ s to examine the effect of tip clearance on the first law and second law performances of the PFHS. The following results were derived from the study:

- The increase in the Re leads to enhance the convective heat transfer coefficient for five studied ξ s. Also, for a specific Re , the increase in the ξ from 1.5 mm to 2.25 mm results in enhancing the convective heat transfer because of the intensification of the vortex strength. However, eliminating the tip clearance, which is obtained by increasing the ξ to 2.5 mm, leads to decrease the convective heat transfer coefficient due to decrease in the vorticity intensification.
- The PFHS with ξ of 2.25 mm requires the highest pumping power with an insignificant discrepancy (0.98%) with other four PFHSs.

- The PFHS with ξ of 1.5 mm has the best performance from the second law of thermodynamic view point due to the minimum entropy generation rates. However, the ξ of 2.25 mm brings the highest performance evaluation criteria ($PEC = 1.28$).
- The shorter ξ (longer tip clearance distances) causes the fluid fastly moves through the PFHS, thereby decreasing the heat transfer time, heat transfer surface area, and uniformity. Thereby, the heat transfer from the PFHS is less uniform in shorter ξ s.

Credit author statement

The main idea of this paper belongs to Amin Shahsavar. The required CDF simulations were performed by Mohammad Shahmohammadi. Ighball Baniasad Askari wrote the article and Amin Shahsavar edited the text.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] J.F. Tullius, T.K. Tullius, Y. Bayazitoglu, Optimization of short micro pin fins in minichannels, *Int. J. Heat Mass Transf.* 55 (2012) 3921–3932.
- [2] R.L. Webb, N.H. Kim, *Principles of Enhanced Heat Transfer*, Taylor & Francis, 2005.
- [3] E. Rasouli, C. Naderi, V. Narayanan, Pitch and aspect ratio effects on single-phase heat transfer through microscale pin fin heat sinks, *Int. J. Heat Mass Transf.* 118 (2018) 416–428.
- [4] J. Zhao, S. Huang, L. Gong, Z. Huang, Numerical study and optimizing on micro square pin-fin heat sink for electronic cooling, *Appl. Therm. Eng.* 93 (2016) 1347–1359.
- [5] S. Ndao, Y. Peles, M.K. Jensen, Effects of pin fin shape and configuration on the single-phase heat transfer characteristics of jet impingement on micro pin fins, *Int. J. Heat Mass Transf.* 70 (2014) 856–863.
- [6] H. Shafeie, O. Abouali, K. Jafarpur, G. Ahmadi, Numerical study of heat transfer performance of single-phase heat sinks with micro pin-fin structures, *Appl. Therm. Eng.* 58 (2013) 68–76.
- [7] P.D. Shima, J. Philip, B. Raj, Influence of aggregation on thermal conductivity in stable and unstable nanofluids, *Appl. Phys. Lett.* 97 (2010) 2008–2011.
- [8] E.M. Sparrow, J.W. Ramsey, Heat transfer and pressure drop for a staggered wall attached array of cylinders with tip clearance, *Int. J. Heat Mass Transf.* 21 (1978) 1369–1377.
- [9] E.M. Sparrow, D.S. Kadle, Effect of tip-to-shroud clearance on turbulent heat transfer from a shrouded, longitudinal fin array, *J. Heat Transf.* 108 (1986) 519–524.
- [10] H. Tabkhi, A. Nayebedeh, Y. Peles, Experimental and numerical local heat transfer study on micro pin fin with tip clearance, *Appl. Therm. Eng.* 179 (2020) 115756.
- [11] R.A. Wirtz, W. Chen, R. Zhou, Effect of flow bypass on the performance of longitudinal fin heat sinks, *J. Electron. Packag.* 116 (2008) 206.
- [12] R.L. Mahajan, Effects of tip clearance and fin density on the performance of heat sinks for VLSI packages, *IEEE Trans. Components Hybrids Manuf. Technol.* 12 (1989) 757–765.
- [13] J.Y. Min, S.P. Jang, S.J. Kim, Effect of tip clearance on the cooling performance of a microchannel heat sink, *Int. J. Heat Mass Transf.* 47 (2004) 1099–1103.
- [14] F. Sastre, A. Valeije, E. Martin, A. Velazquez, International journal of thermal sciences experimental and numerical study on the flow topology of finned heat sinks with tip clearance, *Int. J. Therm. Sci.* 132 (2018) 146–160.
- [15] L. Zhigang, G. Ning, Zh. Chengwu, Influence of tip clearance on heat transfer efficiency in micro-cylinders-group heat sink, *Exp. Thermal Fluid Sci.* 46 (2013) 64–73.
- [16] B. Rajabi Far, S.K. Mohammadian, S.K. Khanna, Y. Zhang, Effects of pin tip-clearance on the performance of an enhanced microchannel heat sink with oblique fins and phase change material slurry, *Int. J. Heat Mass Transf.* 83 (2015) 136–145.
- [17] M. Dogan, M. Sivrioglu, Experimental and numerical investigation of clearance gap effects on laminar mixed convection heat transfer from fin array in a horizontal channel-a conjugate analysis, *Appl. Therm. Eng.* 40 (2012) 102–113.
- [18] J.W. Kim, S.J. Kim, J.I. Min, S.G. Lee, Experimental study on the effect of tip clearance for a straight fin heat sink, *Trans. Korean Soc. Mech. Eng.* (2004) 1189–1194.
- [19] H.K. Park, B.J. Chung, Optimal tip clearance in the laminar forced convection heat transfer of a finned plate in a square duct, *Int. Commun. Heat Mass Transf.* 63 (2015) 73–81.
- [20] M. Reyes, J.R. Arias, A. Velazquez, J.M. Vega, Experimental study of heat transfer and pressure drop in micro-channel based heat sinks with tip clearance, *Appl. Therm. Eng.* 31 (2011) 887–893.
- [21] D. Mei, X. Lou, M. Qian, Zh. Yao, L. Liang, Z. Chen, Effect of tip clearance on the heat transfer and pressure drop performance in the micro-reactor with micro-pin-fin arrays at low Reynolds number, *Int. J. Heat Mass Transf.* 70 (2014) 709–718.
- [22] R. Jadhav, C. Balaji, Fluid flow and heat transfer characteristics of a vertical channel with detached pin-fin arrays arranged in staggered manner on two opposite end walls, *Int. J. Therm. Sci.* 105 (2016) 57–74.
- [23] J.Y. Ho, K.K. Wong, K.C. Leong, T.N. Wong, Convective heat transfer performance of airfoil heat sinks fabricated by selective laser melting, *Int. J. Therm. Sci.* 114 (2017) 213–228.
- [24] K.A. Moores, Y.K. Joshi, Effect of tip clearance on the thermal and hydrodynamic performance of a shrouded pin fin array, *J. Heat Transf.* 125 (2003) 999–1006.
- [25] H. Shen, G. Xie, Ch.Ch. Wang, Thermal performance and entropy generation of novel X-structured double layered microchannel heat sinks, *J. Taiwan Inst. Chem. Eng.* 111 (2020) 90–104.
- [26] T. Ambreen, A. Saleem, C.W. Park, Analysis of hydro-thermal and entropy generation characteristics of nanofluid in an aluminium foam heat sink by employing Darcy-Forchheimer-Brinkman model coupled with multiphase Eulerian model, *Appl. Therm. Eng.* 173 (2020) 115231.
- [27] A.A. Alfaryjat, A. Dobrovicescu, D. Stanciu, Influence of heat flux and Reynolds number on the entropy generation for different types of nanofluids in a hexagon microchannel heat sink, *Chin. J. Chem. Eng.* 27 (2019) 501–513.
- [28] A. Shahsavari, O. Yari, I. Baniasad Askari, The entropy generation analysis of forward and backward laminar water flow in a plate-pin-fin heatsink considering three different splitters, *Int. Commun. Heat Mass Transf.* 120 (2021), 105026.
- [29] A.A.A.A. Al-Rashed, A. Shahsavari, O. Rasooli, M.A. Moghimi, A. Karimipour, M. D. Tran, Numerical assessment into the hydrothermal and entropy generation characteristics of biological water-silver nano-fluid in a wavy walled microchannel heat sink, *Int. Commun. Heat Mass Transf.* 104 (2019) 118–126.
- [30] Zh. Shi, T. Dong, Entropy generation and optimization of laminar convective heat transfer and fluid flow in a microchannel with staggered arrays of pin fin structure with tip clearance, *Energy Convers. Manag.* 94 (2015) 493–504.
- [31] H. Ullah, H.S. Ahmad, M.Sh. Alhodaly, Entropy generation and heat transfer analysis in power-law fluid flow: finite difference method, *Int. Commun. Heat Mass Transf.* 122 (2021), 105111.
- [32] Sh. Hussain, M. Jamal, S.E. Ahmed, Hydrodynamic forces and heat transfer of nanofluid forced convection flow around a rotating cylinder using finite element method: the impact of nanoparticles, *Int. Commun. Heat Mass Transf.* 108 (2019), 104310.
- [33] S.M. Seyyedi, A.S. Dogonchi, M. Hashemi-Tilehnoee, M. Waqas, D.D. Ganji, Investigation of entropy generation in a square inclined cavity using control volume finite element method with aided quadratic Lagrange interpolation functions, *Int. Commun. Heat Mass Transf.* 110 (2020), 104398.
- [34] M.A. Ashraf, Ch. Li, D. Zhang, D. Ross, Retraction notice to “Battery thermal management with conjugate heat transfer in heat sink with Fe₃O₄/CNT-water nanofluid using lattice Boltzmann/finite volume method” [CEP 146 (2019) 107708], *Chem. Eng. Process. Process Intensif.* 155 (2020), 108020.
- [35] R. Mohebbi, A. Amiri Delouei, A. Jamali, M. Izadi, A.A. Mohamad, Pore-scale simulation of non-Newtonian power-law fluid flow and forced convection in partially porous media: thermal lattice Boltzmann method, *Physica A: Stat. Mech. Appl.* 525 (2019) 642–656.
- [36] B. Afra, A. Amiri Delouei, M. Mostafavi, A. Tarokh, Fluid-structure interaction for the flexible filament's propulsion hanging in the free stream, *J. Mol. Liq.* 323 (2021), 114941.
- [37] A. Jalali, A. Amiri Delouei, M. Khorashadizadeh, A.M. Golmohamadi, S. Karimnejad, Mesoscopic simulation of forced convective heat transfer of Carreau-Yasuda fluid flow over an inclined square: temperature-dependent viscosity, *J. Appl. Comput. Mech.* 6 (2) (2020) 307–319.
- [38] A. Shahsavari, M.M. Baseri, A.A.A.A. Al-Rashed, M. Afrand, Numerical investigation of forced convection heat transfer and flow irreversibility in a novel heatsink with helical microchannels working with biologically synthesized water-silver nano-fluid, *Int. Commun. Heat Mass Transf.* 108 (2019) 104324.
- [39] H.C.C. Chiu, R.H.H. Hsieh, K. Wang, J.H.H. Jang, C.R.R. Yu, The heat transfer characteristics of liquid cooling heat sink with micro pin fins, *Int. Commun. Heat Mass Transf.* 86 (2017) 174–180.